

# OOP Lab Project

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***Project Report-  
Academy Management  
System***

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## Description:

The Academy Management System is a C++ program designed to manage an educational academy. It allows the administration, teachers, and students to perform various tasks through a command-line interface. This system consists of four main classes.

## Classes Details:

### 1. Academy Class

- **Purpose:** The Academy class is the entry point of the application. It manages the main execution flow and orchestrates interactions between different user roles.
- **Key Function:**  
void run(): This function manages the main loop of the program, handling user inputs and coordinating actions based on the user role.

### 2. Student Class

The student class contains all the functions and attributes of students including student name and passwords and his subjects and marks in subjects.

- **Attributes:** Username, Passwords, marks, subjects, fees Status.
- **Key Functions:** Functions to return marks and subjects and fees status.

### 3. Teacher Class

The Teacher class contains all information about themselves and can manage students results.

- **Attributes:** Username, Passwords, subjects, salary Status.
- **Key Functions:** Functions to change marks and compile results and see salary status

### 4. Admin Class

- **Attributes:** Admin Username and passwords that are constant.
- **Key Functions:** Functions to add and remove students and teachers and manage fees and salary.

## Classes Workflow:

### 1. Admin Login:

- The admin can log in using a predefined username and password.
- After logging in, the admin can add or remove students and teachers, manage fees, and manage teacher salaries.

## 2. Teacher Login:

- Teachers log in with their credentials.
- They can change student marks, compile student results, and view their salary status.

## 3. Student Login:

- Students log in with their credentials.
- They can view their results and fee status.

## ***OOP Features Used:***

In this AMS we have used following OOP Features:

- **Friend Classes:** To create relationships between classes.
- **Operator Overloading:** To input and display Students and Teachers Details.
- **Data Encapsulation:** To ensure that specific tasks are performed by designated personnel.

## ***Conclusion:***

The Academy Management System is a comprehensive solution for managing students, teachers, and administrative tasks in an educational institution. It ensures that all stakeholders can efficiently perform their roles, contributing to a streamlined management process.